

AI & Data Engineer with production experience across agentic platforms, LLM orchestration, and scalable data infrastructure, specializing in translating complex engineering into real business impact through productionized enterprise AI systems at SAP Labs.

WORK EXPERIENCE

SAP Labs — AI & Data Engineering Intern (Aug 2025 - Present)

SAP BTP | Cloud Integration | Data Engineering Team

- **Architected and productionized agentic AI pipelines** (LangChain, LangGraph, FastAPI) for **Tier-1 enterprise workloads**, ensuring stable inference and platform efficiency **during Black Friday peak loads**.
- **Introduced** and owned a **first-of-its-kind** (within team) **MCP integration layer** unifying monitoring systems and data pipelines; **led end-to-end development** of an **MCP code execution PoC proving ~88% token reduction** and **~57%+ cost savings** with improved latency — presented at internal SAP forums, sparking interest from global teams.
- **Monitored real-time telemetry across high-throughput multi-tenant observability systems** (~70% Fortune 500 customers), detecting cross-pipeline anomalies and ensuring production reliability.
- **Diagnosed and resolved Kubernetes cluster instabilities** (resource contention, scheduling inefficiencies, service failures), improving system stability and maintaining high availability.
- **Partnered** with lead architect, and data scientists to **design and build a data pipeline monitoring** system on **HDLFS** — spanning **Kibana ingestion** through **bronze-to-silver/gold layer scripts**, deployed on **Databricks+Spark Compute** — **automating critical data gap detection and alerting, reducing manual monitoring** overhead.

PROJECTS

Model Context Protocol (MCP) Framework for Agentic & LLM Orchestration Systems (IEEE 2026 : Awaiting Publication)

- **Designed a centralized MCP orchestration layer** exposing datasets as callable resources and analytical logic as tools for protocol-mediated execution across agent workflows.
- **Decoupled agent-tool interaction architecture**, replacing tightly coupled integrations with standardized discovery and invocation semantics to enable modular, extensible systems.
- **Built a stateful execution framework enabling multi-stage LLM reasoning, tool chaining, and orchestrated workflows** for scalable agent-driven processing.

OEM Supply Chains: Forecasting, Supplier Intelligence, and Logistics Systems

- **Developed a hybrid demand forecasting stack** (ARIMA, Random Forest, Gradient Boosting, LSTM) powering enterprise supply chain intelligence pipelines.
- **Modeled multi-tier supplier networks as graphs**, applying **GraphSAGE-based routing optimization** for adaptive logistics planning.
- **Engineered an AI-driven decision-support system** integrating forecasting, sourcing, and distribution models for coordinated inventory and supplier optimization.

TECHNICAL EXPERTISE

Languages: Python, SQL

AI Stack: LangChain, LangGraph, MCP, FastAPI, Prompt Engineering, Agentic Architectures

ML & Deep Learning: TensorFlow, PyTorch, Computer Vision, ML Modelling, Ensemble Methods

Data Science & Analytics: Feature Engineering, Data Visualization, Metabase, Statistics, Predictive Modeling

Databases: PostgreSQL, SAP HANA, MongoDB, SQL Systems

Platforms & Infrastructure: SAP BTP, Cloud Foundry, Kubernetes, Distributed/Microservices Systems, Telemetry

Full-Stack: Node.js, JavaScript, HTML, CSS, React, Streamlit

Developer Tools: Git, Databricks, Jira, VS Code, Jupyter

ACHIEVEMENTS

SAP Recognition Award (Monetary) — Dec 2025

Recognized by product leadership for engineering contributions to productionized VIP Agent adopted by SRE and L1/L2 operations teams.

Citibank Ada Lovelace Hackathon 2024

Finalist (4th Place) in the Citibank Hackathon held @ Citi IT HQ Chennai, Taramani

Volkswagen i.mobliothon 4.0 Hackathon

Semi-Finalist (National Top-50) @ Volkswagen Hackathon 2024 - Received PPO opportunity from Volkswagen for engineering solution.

The World Scholar's Cup - Global Round 2019

Secured 3 medals across leadership, analytical writing, and competitive debate @ Worlds scholars cup Manila, Philippines

EDUCATION

SRM Institute of Science and Technology
Chennai

Vels University
Chennai

M.tech (Integrated) Computer Science Engineering with
Specialization in Data Science (CGPA 9.15) 2021 – 2026

12th Grade Central Board of Secondary Education - 92 %
2021

